Newton’s Laws of Motion (p270-273)

Newton’s First Law

1. For thousands of years, why did people believe objects stopped moving?

2. What did Galileo realise and then reason?

3. What is Newton’s first law of motion?

4. What is inertia and how is mass related to inertia?

5. Describe an example of Newton’s first law.

6. Describe what happens in a car accident and why seatbelts are important.

Newton’s Second Law

1. What is Newton’s second law?

2. Write the equation that expresses Newton's second law. State what each symbol represents.

3. Describe an example of Newton's second law.

4. Calculate the force needed to accelerate a 2kg box at 2 m/s^2. Show your working.

5. Calculate the mass, in kilograms, of a crate which is pushed by a force of 45N and accelerates at 0.05m/s^2.

6. A 70kg runner exerts a force of 56N. What is the acceleration of the runner?

Newton's Third Law

1. What did Newton realise?

2. What is Newton's third law?

3. Give two examples of pairs of action-reaction forces.